

Skill 3

```
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE skill3_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE skill3_db;CREATE TABLE customers (
Database changed
->   customer_id INT PRIMARY KEY,
->   customer_name VARCHAR(100),
->   city VARCHAR(50)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE accounts (
->   account_id INT PRIMARY KEY,
->   customer_id INT,
->   account_type VARCHAR(30),
->   balance DECIMAL(10,2),
->   FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE transactions (
->   txn_id INT PRIMARY KEY,
->   account_id INT,
->   txn_type VARCHAR(20),
->   amount DECIMAL(10,2),
->   txn_date DATE,
->   FOREIGN KEY (account_id) REFERENCES accounts(account_id)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE customer_details (
->   customer_id INT PRIMARY KEY,
->   details JSON
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO customers VALUES
-> (1, 'Ravi', 'Hyderabad'),
-> (2, 'Sita', 'Vizag'),
-> (3, 'Kiran', 'Vijayawada'),
-> (4, 'Anil', 'Hyderabad'),
-> (5, 'Priya', 'Vizag');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO accounts VALUES
-> (101, 1, 'Savings', 50000),
-> (102, 2, 'Savings', 35000),
-> (103, 3, 'Current', 80000),
-> (104, 4, 'Savings', 45000),
-> (105, 5, 'Current', 60000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO transactions VALUES
-> (1001, 101, 'Deposit', 5000, '2026-01-05'),
-> (1002, 101, 'Withdrawal', 2000, '2026-01-10'),
-> (1003, 102, 'Deposit', 7000, '2026-01-12'),
-> (1004, 103, 'Deposit', 10000, '2026-01-15'),
-> (1005, 104, 'Withdrawal', 3000, '2026-01-18'),
-> (1006, 105, 'Deposit', 8000, '2026-01-20'),
-> (1007, 102, 'Withdrawal', 1500, '2026-01-22'),
-> (1008, 103, 'Withdrawal', 4000, '2026-01-25'),
-> (1009, 104, 'Deposit', 6000, '2026-01-27'),
-> (1010, 105, 'Withdrawal', 2500, '2026-01-30');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
```

```

mysql>
mysql> INSERT INTO customer_details VALUES
-> (1, '{"phone":"9876543210","email":"ravi@example.com","status":"Active"}'),
-> (2, '{"phone":"9876543211","email":"sita@example.com","status":"Active"}'),
-> (3, '{"phone":"9876543212","email":"kiran@example.com","status":"Active"}'),
-> (4, '{"phone":"9876543213","email":"anil@example.com","status":"Inactive"}'),
-> (5, '{"phone":"9876543214","email":"priya@example.com","status":"Active"}');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM customers;
+-----+-----+-----+
| customer_id | customer_name | city |
+-----+-----+-----+
| 1 | Ravi | Hyderabad |
| 2 | Sita | Vizag |
| 3 | Kiran | Vijayawada |
| 4 | Anil | Hyderabad |
| 5 | Priya | Vizag |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> INSERT INTO customers
-> VALUES (6, 'Rahul', 'Guntur');
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM customers
-> WHERE customer_id = 6;
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE accounts
-> SET balance = balance + 5000
-> WHERE account_id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM customers
-> WHERE customer_id = 6;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT

mysql> SELECT
-> c.customer_name,
-> a.account_id,
-> a.account_type,
-> a.balance
-> FROM customers c
-> INNER JOIN accounts a
-> ON c.customer_id = a.customer_id;
+-----+-----+-----+-----+
| customer_name | account_id | account_type | balance |
+-----+-----+-----+-----+
| Ravi | 101 | Savings | 55000.00 |
| Sita | 102 | Savings | 35000.00 |
| Kiran | 103 | Current | 80000.00 |
| Anil | 104 | Savings | 45000.00 |
| Priya | 105 | Current | 60000.00 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT
-> c.customer_name,
-> a.account_id,
-> a.balance
-> FROM customers c
-> LEFT JOIN accounts a
-> ON c.customer_id = a.customer_id;
+-----+-----+-----+
| customer_name | account_id | balance |
+-----+-----+-----+
| Ravi | 101 | 55000.00 |
| Sita | 102 | 35000.00 |
| Kiran | 103 | 80000.00 |
| Anil | 104 | 45000.00 |
| Priya | 105 | 60000.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT
-> c.customer_name,
-> a.account_type

```

```
mysql> SELECT
->     c.customer_name,
->     a.account_type
-> FROM customers c
-> CROSS JOIN accounts a;
```

customer_name	account_type
Priya	Savings
Anil	Savings
Kiran	Savings
Sita	Savings
Ravi	Savings
Priya	Savings
Anil	Savings
Kiran	Savings
Sita	Savings
Ravi	Savings
Priya	Current
Anil	Current
Kiran	Current
Sita	Current
Ravi	Current
Priya	Savings
Anil	Savings
Kiran	Savings
Sita	Savings
Ravi	Savings
Priya	Current
Anil	Current
Kiran	Current
Sita	Current
Ravi	Current

25 rows in set (0.00 sec)

```
mysql> SELECT customer_name
-> FROM customers
-> WHERE customer_id IN (
->     SELECT customer_id
```

```

mysql> SELECT customer_name
-> FROM customers
-> WHERE customer_id IN (
->     SELECT customer_id
->     FROM accounts
->     WHERE balance > (
->         SELECT AVG(balance)
->         FROM accounts
->     )
-> );
+-----+
| customer_name |
+-----+
| Kiran         |
| Priya         |
+-----+
2 rows in set (0.00 sec)

mysql> WITH HighBalance AS (
->     SELECT customer_id, balance
->     FROM accounts
->     WHERE balance > 50000
-> )
-> SELECT
->     c.customer_name,
->     h.balance
-> FROM HighBalance h
-> JOIN customers c
-> ON h.customer_id = c.customer_id;
+-----+-----+
| customer_name | balance |
+-----+-----+
| Ravi          | 55000.00 |
| Kiran         | 80000.00 |
| Priya         | 60000.00 |
+-----+-----+
3 rows in set (0.00 sec)

```

```
mysql> SELECT
->     account_type,
->     COUNT(*) AS total_accounts,
->     AVG(balance) AS average_balance
-> FROM accounts
-> GROUP BY account_type
-> HAVING AVG(balance) > 40000;
+-----+-----+-----+
| account_type | total_accounts | average_balance |
+-----+-----+-----+
| Savings      | 3              | 45000.000000    |
| Current      | 2              | 70000.000000    |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT
->     account_id,
->     customer_id,
->     balance,
->     RANK() OVER (ORDER BY balance DESC) AS balance_rank
-> FROM accounts;
+-----+-----+-----+-----+
| account_id | customer_id | balance | balance_rank |
+-----+-----+-----+-----+
| 103        | 3           | 80000.00 | 1             |
| 105        | 5           | 60000.00 | 2             |
| 101        | 1           | 55000.00 | 3             |
| 104        | 4           | 45000.00 | 4             |
| 102        | 2           | 35000.00 | 5             |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT
->     account_id,
->     balance,
->     CASE
->         WHEN balance >= 70000 THEN 'High Balance'
->         WHEN balance >= 40000 THEN 'Medium Balance'
->         ELSE 'Low Balance'
->     
```

```
mysql> SELECT customer_name, city
-> FROM customers
-> WHERE city = 'Hyderabad'
->
-> UNION
->
-> SELECT customer_name, city
-> FROM customers
-> WHERE city = 'Vizag';
+-----+-----+
| customer_name | city |
+-----+-----+
| Ravi          | Hyderabad |
| Anil          | Hyderabad |
| Sita          | Vizag     |
| Priya         | Vizag     |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT customer_id
-> FROM accounts
-> WHERE balance > 40000
->
-> INTERSECT
->
-> SELECT customer_id
-> FROM customers
-> WHERE city = 'Hyderabad';
+-----+
| customer_id |
+-----+
| 1           |
| 4           |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT customer_id
-> FROM customers
->
->
```

```
->
-> SELECT customer_id
-> FROM accounts;
Empty set (0.00 sec)
```

```
mysql> SELECT
->     customer_id,
->     JSON_UNQUOTE(JSON_EXTRACT(details, '$.email')) AS email
-> FROM customer_details;
```

customer_id	email
1	ravi@example.com
2	sita@example.com
3	kiran@example.com
4	anil@example.com
5	priya@example.com

5 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM customer_details
-> WHERE JSON_UNQUOTE(JSON_EXTRACT(details, '$.status')) = 'Active';
```

customer_id	details
1	{"email": "ravi@example.com", "phone": "9876543210", "status": "Active"}
2	{"email": "sita@example.com", "phone": "9876543211", "status": "Active"}
3	{"email": "kiran@example.com", "phone": "9876543212", "status": "Active"}
5	{"email": "priya@example.com", "phone": "9876543214", "status": "Active"}

4 rows in set (0.00 sec)

```
mysql> |
```